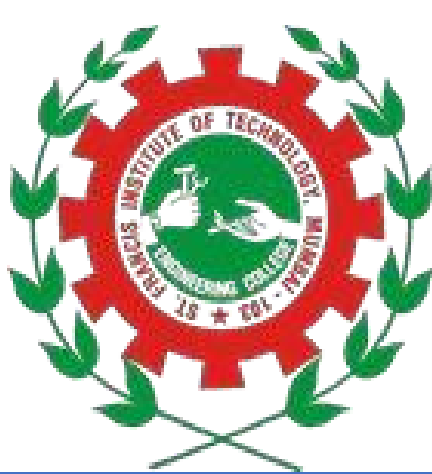




Title of the Project : Secure Suite AI

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Abstract

SecureAI Suite is an integrated AI-driven cybersecurity platform developed to automate vulnerability detection, phishing URL classification, subdomain discovery, and malware analysis. It combines static and machine learning-based scanners to deliver real-time, actionable threat insights. Using Random Forest and rule-based modules, it identifies risks and suggests intelligent fixes. The unified dashboard ensures faster decision-making and improved protection against evolving cyber threats.

Problem Statement

Existing cybersecurity solutions operate in isolation, leading to delayed and inefficient threat response. This project aims to develop and implement a unified AI-based security suite that automates vulnerability scanning, phishing detection, and malware analysis. The objective is to centralize threat intelligence and provide real-time automated remediation to strengthen overall system defense.

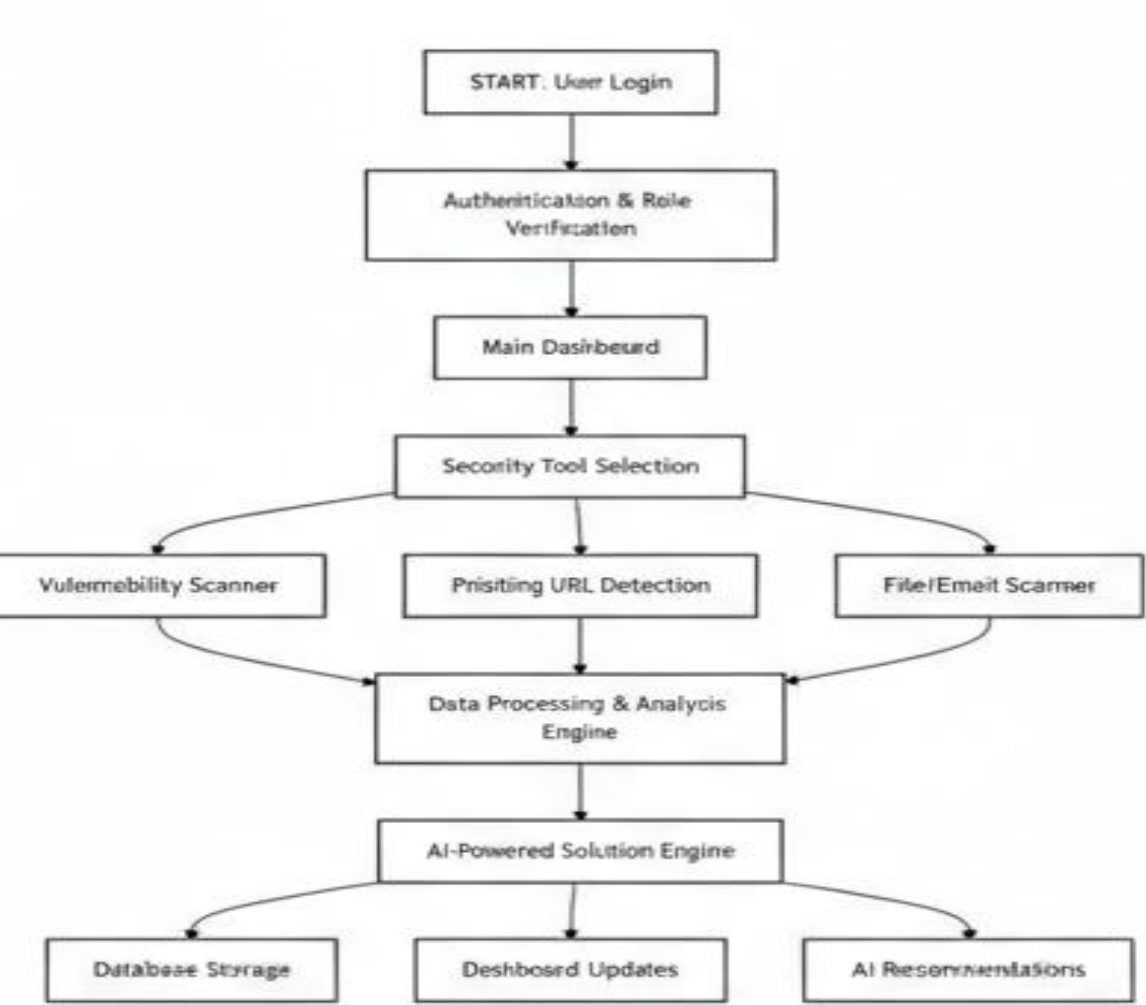


Figure 1: Workflow diagram

Review of Literature

Title of Paper	Methodology	Discussion of results	Drawback
PhishHaven: An Efficient Real-Time AI Phishing URLs Detection System (2020)	Combined several machine learning models to detect phishing URLs in real-time using URL analysis and voting.	Successfully identified most phishing URLs with high accuracy and fast response.	May not detect new, sophisticated phishing URLs as effectively.
Machine Learning for Multiple Stage Phishing URL Prediction (2021)	Evaluated different machine learning algorithms to classify phishing URLs in multiple stages.	Multi-stage approach and logistic regression improved phishing detection rates.	Reliance on dataset features may impact detection of future trends.
A Comprehensive Vulnerability Tools Analysis (2024)	Compared widely used vulnerability tools in organizational security	Integrating these tools resulted in more effective vulnerability management.	Implementation requires expertise and significant resources.

Proposed Methodology

SecureAI Suite integrates four main modules: Vulnerability Scanner, Subdomain Finder, Phishing URL Detector, and Malware File Scanner. The scanner module uses tools like pip-audit, npm audit, and OSV-Scanner to identify CVEs and recommend fixes. The phishing detector employs a Random Forest classifier on extracted URL features, while the malware module analyzes file entropy and opcode patterns. All modules interact through a FastAPI backend and a unified Next.js dashboard for monitoring and control.

Experimental setup

The platform is developed using FastAPI in Python for the backend, Next.js and TailwindCSS for the frontend (deployed on Vercel), and MongoDB for data storage. Machine learning models are trained in Jupyter Notebook using scikit-learn with datasets from PhishTank, Kaggle phishing URLs, and EMBER malware datasets. Scanning modules rely on official vulnerability databases such as PyPI, npm advisories, OSV, and CIRCL CVE. APIs are deployed on Render for seamless integration between components.

Results

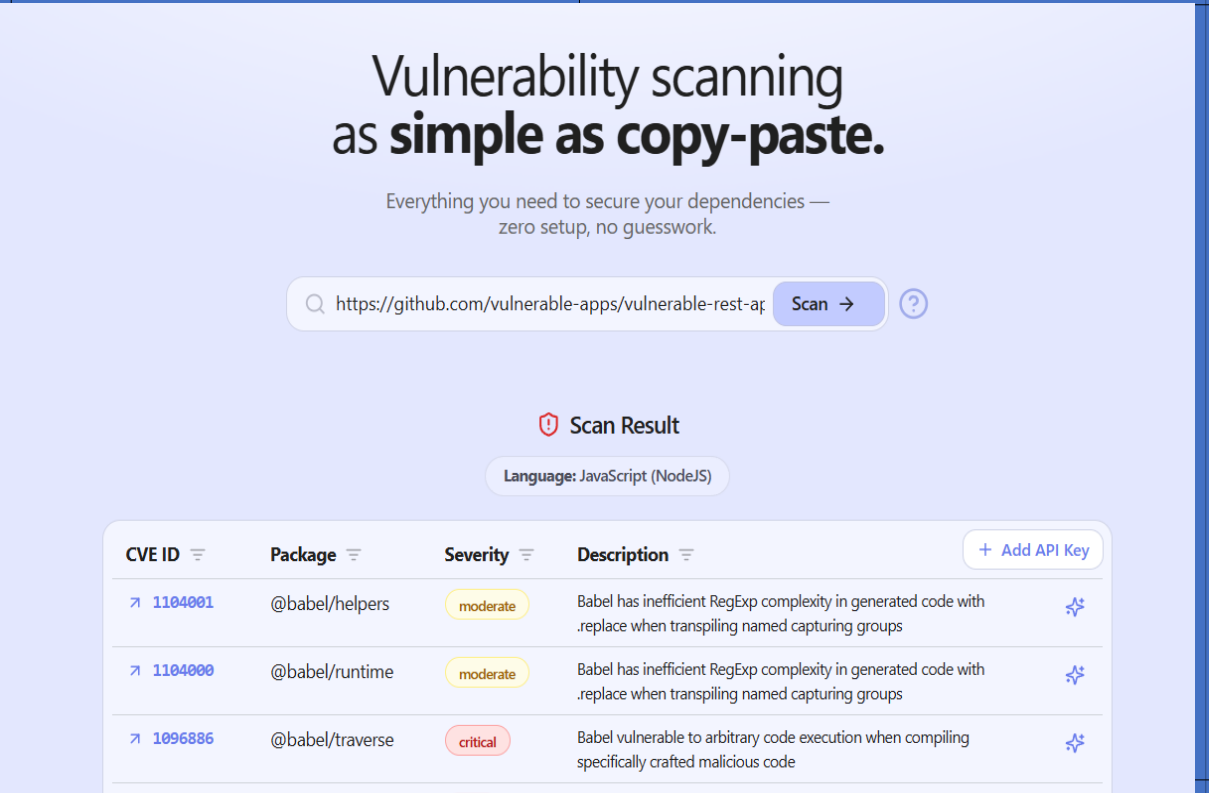


Figure 2: Results

Conclusion and future directions

Conclusion: Secure suite AI is a multi-model approach achieves high accuracy in finding real-time github issues, phishing URL detection, outperforming previous methods and efficiently spotting both human and AI-generated threats.
Future Directions: Enhancing Secure suite ai with multiple cyber security tools, feature optimization, and faster, multi-threaded classification can further boost scalability and detection accuracy.*

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